```
ZooKeeper JMX enabled by default
Using config: /home/user/Desktop/zookeeper-3.4.6/bin/../conf/
zoo.cfg
Starting zookeeper... STARTED
```

c-1) Connecting to the Zookeeper server
Next, give the following command:

```
sudo bin/zkCli.sh
```

You will be connected to the Zookeeper server and will get the following response:

```
Connecting to localhost:2181
2018-05-31 16:48:40,828 [myid:] - INFO  [main:Environment@100]
- Client environment: zookeeper.version=3.4.10-39d3a4f269333c9
22ed3db283be479f9deacaa0f, built on 03/23/2017 10:13 GMT
2018-05-31 16:48:40,846 [myid:] - INFO  [main:Environment@100]
- Client environment:host.name=linux-neethu
2018-05-31 16:48:40,846 [myid:] - INFO  [main:Environment@100]
- Client environment:java.version=1.8.0_171
---------------------------
Welcome to ZooKeeper!
----------------------------
WATCHER::
WatchedEvent state:SyncConnected type:None path:null
[zk: localhost:2181(CONNECTED) 0]
```

d) Stopping the Zookeeper server
After connecting the server and performing all the operations, we can stop the Zookeeper server with the following command:

```
$ sudo bin/zkServer.sh stop
```

Step 2: Apache Kafka installation

Perform the following steps to install Kafka on your machine.

a) Downloading Kafka and extracting the tar file
To install Kafka on your machine, click on *https://www.apache.org/dyn/closer.cgi?path=/kafka/0.9.0.0/kafka_2.11-0.9.0.0.tgz.* Extract the *tar* file using the following command:

```
 tar -zxf kafka_2.11-1.1.0 tar.gz
cd kafka_2.11-1.1.0/
```

b) Starting the server
Start the server by using the following commands:

```
bin/kafka-server-start.sh config/server.properties
 [2018-05-31 11:53:48,631] INFO KafkaConfig values:
            alter.log.dirs.replication.quota.window.size.
seconds = 1
 broker.id.generation.enable = true
 log.preallocate = false
 log.roll.hours = 168
 security.inter.broker.protocol = PLAINTEXT
 request.timeout.ms = 30000
 zookeeper.connect = localhost:2181
…………………………………….
```

c) Stopping the server
Use the following command to stop the Kafka server:

```
 bin/kafka-server-stop.sh config/server.properties
```

Single node, single broker configuration

Now that we have installed Zookeeper and Kafka on our machine, start Zookeeper before moving to the Kafka cluster.

Open a new terminal and type the following command:

```
bin/zookeeper-server-start.sh config/zookeeper.properties
```

To start Kafka broker, use the following command:

```
bin/kafka-server-start.sh config/server.properties
```

In this configuration, we have a single Zookeeper and broker ID instance. The following steps will help you configure it.

a) Creating a Kafka topic
Topics are logical groupings of messages and Kafka provides a command line utility named *kafka-topics.sh* to create a topic on the server. Open a new terminal and type the following example:

```
bin/kafka-topics.sh --create --zookeeper localhost:2181
--replication-factor 1
--partitions 1 --topic topic-name
Example
bin/kafka-topics.sh --create --zookeeper localhost:2181
--replication-factor 1
--partitions 1 --topic Hello-Kafka
```

We just created a topic named *Hello-Kafka* with a single partition and one replica factor, similar to the following output:

```
Output – Created topic Hello-Kafka
```

Once the topic has been created, we will get a notification in the Kafka broker terminal window and the log for the created topic specified in */tmp/kafka-logs/* in the *config/server.properties* file.

b) Starting producers to send messages
Use the command given below:

```
bin/kafka-console-producer.sh --broker-list localhost:9092
--topic Hello-Kafka
```

Continued on page 46...